Solar Energy: How Photovoltaic Power Works

Why Solar Energy Matters

The Sun delivers far more energy to the Earth's surface in a single hour than humanity uses in an entire year. Capturing even a small fraction of this energy can replace electricity generated from coal, oil and natural gas, which release carbon dioxide and air pollutants. Solar power has become one of the fastest-growing sources of electricity in the world, largely because the cost of solar panels has fallen dramatically, by roughly 90 percent between 2010 and 2020. In many countries, including India, new utility-scale solar plants now produce some of the cheapest electricity available.

There are two main ways to turn sunlight into useful energy. Solar thermal systems use sunlight to heat water or another fluid, either for direct use such as household water heaters, or to drive a steam turbine in concentrated solar power plants. Photovoltaic (PV) systems convert sunlight directly into electricity using semiconductor devices called solar cells. This document focuses on photovoltaic technology, which accounts for the vast majority of solar electricity generated today.

The Photovoltaic Effect

The photovoltaic effect was first observed by the French physicist Edmond Becquerel in 1839, when he noticed that certain materials produced a small electric current when exposed to light. The first practical silicon solar cell was demonstrated at Bell Laboratories in 1954 by Daryl Chapin, Calvin Fuller and Gerald Pearson, with an efficiency of about 6 percent. Early solar cells were very expensive and were used mainly to power satellites, beginning with Vanguard 1 in 1958.

A solar cell is made from a semiconductor, most commonly crystalline silicon. The silicon is doped to create two layers: an n-type layer with extra electrons and a p-type layer with extra holes, where electrons are missing. Where the two layers meet, a p-n junction forms, and an internal electric field is created. When a photon of sunlight with enough energy strikes the cell, it knocks an electron loose and creates an electron-hole pair. The electric field pushes the electrons towards one side and the holes towards the other. Metal contacts on the front and back of the cell collect these charges, and when an external circuit is connected, a direct current (DC) flows.

Types of Solar Panels

Monocrystalline panels are made from cells cut from a single, pure crystal of silicon. They have a uniform dark appearance and the highest efficiency of common commercial panels, typically between 20 and 23 percent, but they cost more to produce. Polycrystalline panels are made from silicon fragments melted together. They have a blue, speckled look and somewhat lower efficiency, typically 15 to 18 percent, and have largely been displaced by cheaper monocrystalline production. Thin-film panels are made by depositing very thin layers of photovoltaic material, such as cadmium telluride or amorphous silicon, onto glass, metal or plastic. They are lightweight and flexible and perform relatively well in high temperatures, but they generally have lower efficiency and need more area for the same power.

Newer designs continue to improve performance. PERC cells add a reflective layer at the back of the cell to capture more light. Bifacial panels can collect light on both their front and rear surfaces, gaining extra energy from light reflected off the ground. Perovskite materials and perovskite-silicon tandem cells have achieved very high efficiencies in laboratories and are an active area of research.

Components of a Solar Power System

A complete PV system needs more than panels. Solar panels, also called modules, are made of many cells connected together and generate DC electricity. An inverter converts that DC electricity into alternating current (AC), which is what household appliances and the electricity grid use. String inverters serve a whole row of panels, while microinverters are fitted to each individual panel and reduce losses when some panels are shaded. Mounting structures hold the panels at the correct angle and orientation; in India, panels are usually tilted towards the south at an angle close to the site's latitude. Other components include cables, protection devices such as fuses and surge protectors, a meter, and optionally a battery to store energy for use at night.

Grid-Tied, Off-Grid and Hybrid Systems

A grid-tied system is connected to the electricity grid. During the day, it supplies power to the building, and any surplus is exported to the grid. Under net metering, the electricity utility records both import and export and the consumer is billed only for the net energy used. A grid-tied system without batteries switches off during a power cut for the safety of line workers. An off-grid system is not connected to the grid at all and relies on batteries to supply power at night and on cloudy days; it is useful in remote villages without a reliable grid connection. A hybrid system combines grid connection with battery storage, so that it can export surplus energy and also provide backup power during outages.

Factors Affecting Output

The amount of electricity a panel produces depends on the solar irradiance, which is the power of sunlight reaching a surface and is about 1,000 watts per square metre on a clear day at noon. Panels are rated in watts peak under standard test conditions. Higher temperatures reduce efficiency, because a silicon cell loses roughly 0.3 to 0.5 percent of its output for every degree Celsius above 25 degrees. Shading of even a small part of a panel can significantly reduce the output of a whole string. Dust and dirt also block light, so regular cleaning is important, especially in dry and dusty regions. As a rough guide, in much of India each kilowatt of installed rooftop solar capacity generates about 4 to 5 units (kilowatt-hours) of electricity per day on average.

Solar Energy in India

India has set an ambitious target of 500 gigawatts of non-fossil electricity capacity by 2030, with solar power expected to contribute the largest share. The country has built some of the world's largest solar parks, such as the Bhadla Solar Park in Rajasthan, and has rapidly expanded rooftop solar through government schemes. The PM Surya Ghar: Muft Bijli Yojana, launched in 2024, offers subsidies to help households install rooftop solar systems. India also co-founded the International Solar Alliance with France in 2015

to promote solar energy in sunny countries.

Advantages and Limitations

Solar power produces no direct emissions during operation, uses no fuel, requires little maintenance and has a long service life, as most panels carry performance warranties of 25 years. It can be installed at many scales, from a single rooftop to a large solar park. However, solar output is variable: it stops at night and falls during cloudy weather and the monsoon. Storing energy in batteries adds cost. Large ground-mounted plants need significant land, and panels must be recycled responsibly at the end of their life to recover glass, aluminium, silicon and silver.